# reaction-box-appendix

## 10. The Reaction Box Is a Familiar Quantum Gate in Disguise

Once the box is framed as a control bit plus two reversible input/output qubits, it turns out to be something already studied: the priority bit selects between the two flavors of the **CNOT** gate. The Coylean map is, on this reading, a 2D lattice of CNOTs with a classical control pattern.

### 10.1 The gate

Working from the truth tables in §3, the box implements a unitary `U_P : |v⟩|h⟩ → |d⟩|r⟩` parameterized by the classical priority bit `P`:

```
    Priority V (vertical wins):   d = v,       r = h ⊕ v
    Priority H (horizontal wins): d = v ⊕ h,   r = h
```

In the computational basis `(|00⟩, |01⟩, |10⟩, |11⟩)`:

```
              ┌         ┐                ┌         ┐
              │ 1 0 0 0 │                │ 1 0 0 0 │
    U_V =     │ 0 1 0 0 │        U_H =   │ 0 0 0 1 │
              │ 0 0 0 1 │                │ 0 0 1 0 │
              │ 0 0 1 0 │                │ 0 1 0 0 │
              └         ┘                └         ┘
```

These are precisely the two CNOTs, distinguished only by which qubit plays the control role:

- `U_V = CNOT(v → h)` — `v` is control, `h` gets XORed.
- `U_H = CNOT(h → v)` — `h` is control, `v` gets XORed.

The "XOR toggle with priority breaking ties" rule, written out classically in §3, is literally the quantum CNOT. "Vertical wins" / "horizontal wins" just selects the control/target assignment.

## 10.2 The full controlled gate

Lifting the priority into a quantum register gives a 3-qubit gate `U` on `|P⟩|v⟩|h⟩` whose action is a direction-switching CNOT. A clean decomposition is a Fredkin–CNOT–Fredkin sandwich:

```
P ────●────────●──────
      │        │
v ────×────●───×──────
      │    │   │
h ────×────⊕───×──────
     FRED  CNOT FRED
   (P,v,h) v→h (P,v,h)
```

The outer Fredkins (controlled-SWAPs on `v` and `h`, conditioned on `P`) swap the two data qubits when `P = 1`. Sandwiched around a fixed-direction `CNOT(v → h)`, the net effect is a CNOT whose direction is dictated by `P`: identity-on-direction when `P = 0`, flipped when `P = 1`. Verifying on the basis states reproduces the §3 tables.

---

## 10.3 Reading "4 reversible input/output pairs"

The phrase admits two readings, both pointing at the same physical fact.

**Reading A — 4 ports.** The box has four physical ports (N, W, S, E). Because the gate is involutive, each port carries one bit that can be read in either direction; the labels "input" and "output" are conventional, not physical. Four ports, four bidirectional signals.

**Reading B — 4 basis-state pairings.** The two-qubit gate is a permutation in $S_4$. For each priority it is a product of disjoint 2-cycles (fixed points plus transpositions) — i.e. four reversible input/output state pairings.

Both readings reduce to the single fact $U_P^2 = I$. The gate is unitary **and** Hermitian — a **quantum reflection**.

## 10.4 Why this matters

- A Hermitian unitary is the strongest form of self-inverse gate. The only single-qubit Hermitian unitaries are `I, X, Y, Z, H` and their signed variants. The CNOT is the canonical 2-qubit Hermitian unitary. The reaction box sitting exactly on this point of gate space is not coincidence — it is **what makes seam ↔ patch invertibility hold**. The Coylean property in §4 is literally the Hermitian-unitary property in the 2-qubit setting.
- The Coylean map is a **classical-controlled CNOT lattice**: a 2D tiling of CNOTs with the direction at each cell selected by the 2-adic valuation of the absolute coordinates. The "priority field" of §5 is a classical control pattern over a CNOT lattice.
- This reframes the §7 measures in standard quantum terms. Bijectivity (§7.1) is unitarity. Involutivity (§7.2) is Hermiticity. The Hamming-distance and information-theoretic measures (§7.3, §7.4) are the natural relaxations: distance from a Hermitian unitary in operator norm, and the gate's classical channel capacity.
- The natural generalizations in §6 acquire concrete quantum names. Port arity `n = 3` over `{0,1}` with the strongest involutive candidate is the **Toffoli** gate. Higher-arity / multi-control Hermitian unitaries (controlled-Z, controlled-controlled-NOT, the full Clifford-group involutions) are the gate-level versions of the Coylean family. The search for "other algorithms with this symmetry" becomes the search for **classically-controlled lattices of Hermitian Clifford gates with a non-trivial priority assignment.**

Read this way, the Coylean map is the simplest non-trivial point in a large space already mapped out by reversible-computing and quantum-gate taxonomies — but combined with a number-theoretic priority assignment that those taxonomies do not, on their own, supply. The interesting object is the *combination*: the gate (Hermitian unitary) and the classical control field (2-adic valuation) together.